

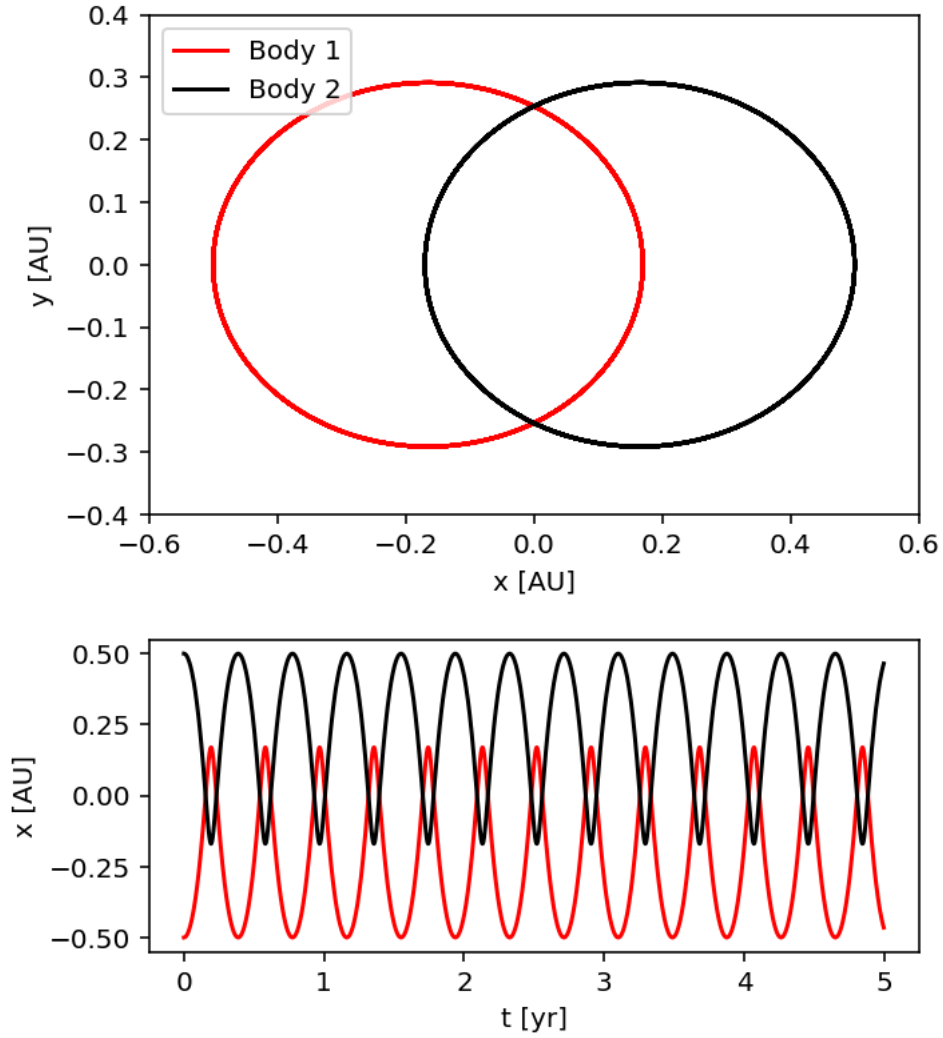


Figure 1: This figure shows the validated two-body simulation for two equal-mass stars orbiting their common centre of mass. The upper panel plots the trajectories in the x-y plane, where both bodies follow closed circular orbits as expected from the analytic two-body solution. Body 1 is shown in red and Body 2 in black. The lower panel shows the x-position of each body over five years, confirming the correct periodic motion and opposite phase of the two stars. This result demonstrates that the force calculation and ODE solver are working correctly and provides the baseline validation before extending the model to three or more bodies

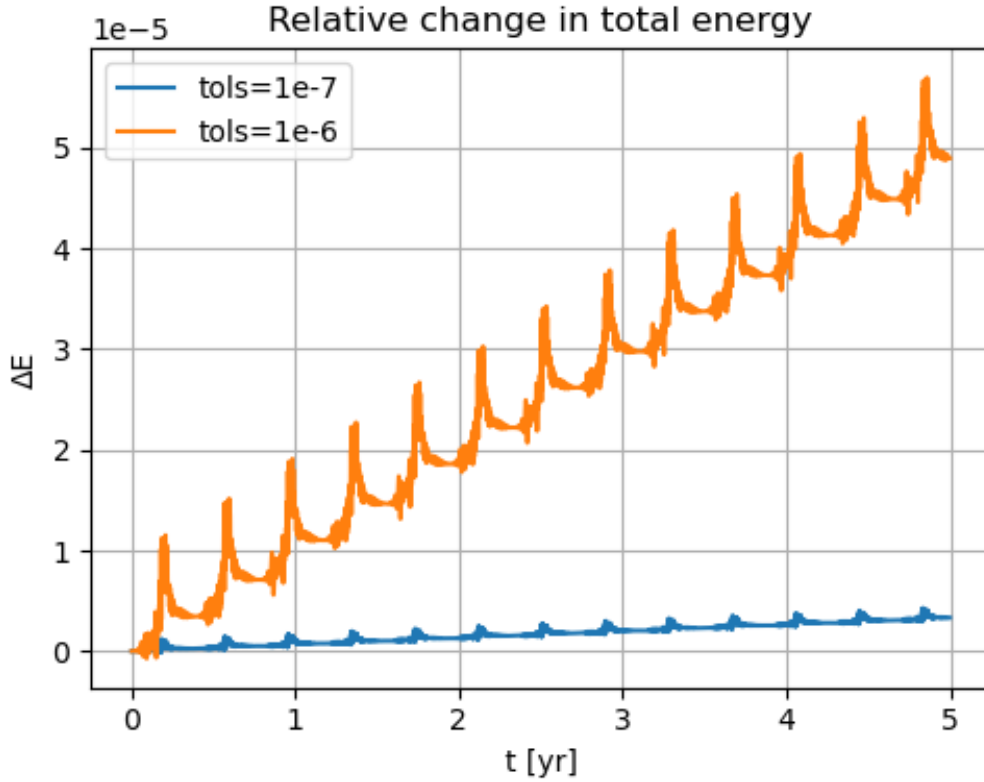


Figure 2: This figure shows the relative change in total energy, $\Delta E = \frac{[E(t)-E(0)]}{E(0)}$, for the equal mass two body orbit over a five-year integration, demonstrating the effect of the ODE solver tolerances on energy conservation. The orbits are integrated with *solve_ivp* using absolute and relative tolerances of 10^{-7} (blue curve) and 10^{-6} (orange curve). In both cases the total energy remains very nearly constant, however the looser tolerance exhibits a systematic growth, reaching $|\Delta E| \approx 5 \times 10^{-5}$ by the end of the run. In contrast, the tighter tolerance keeps $|\Delta E| \lesssim 3 \times 10^{-6}$, more than an order of magnitude smaller. The repeated spikes correspond to each orbital period when the stars pass closest together and the gravitational acceleration is largest, which is the most demanding part of the integration. This test justifies the choice of 10^{-7} tolerances for subsequent simulations as a good compromise between accuracy and computational cost.

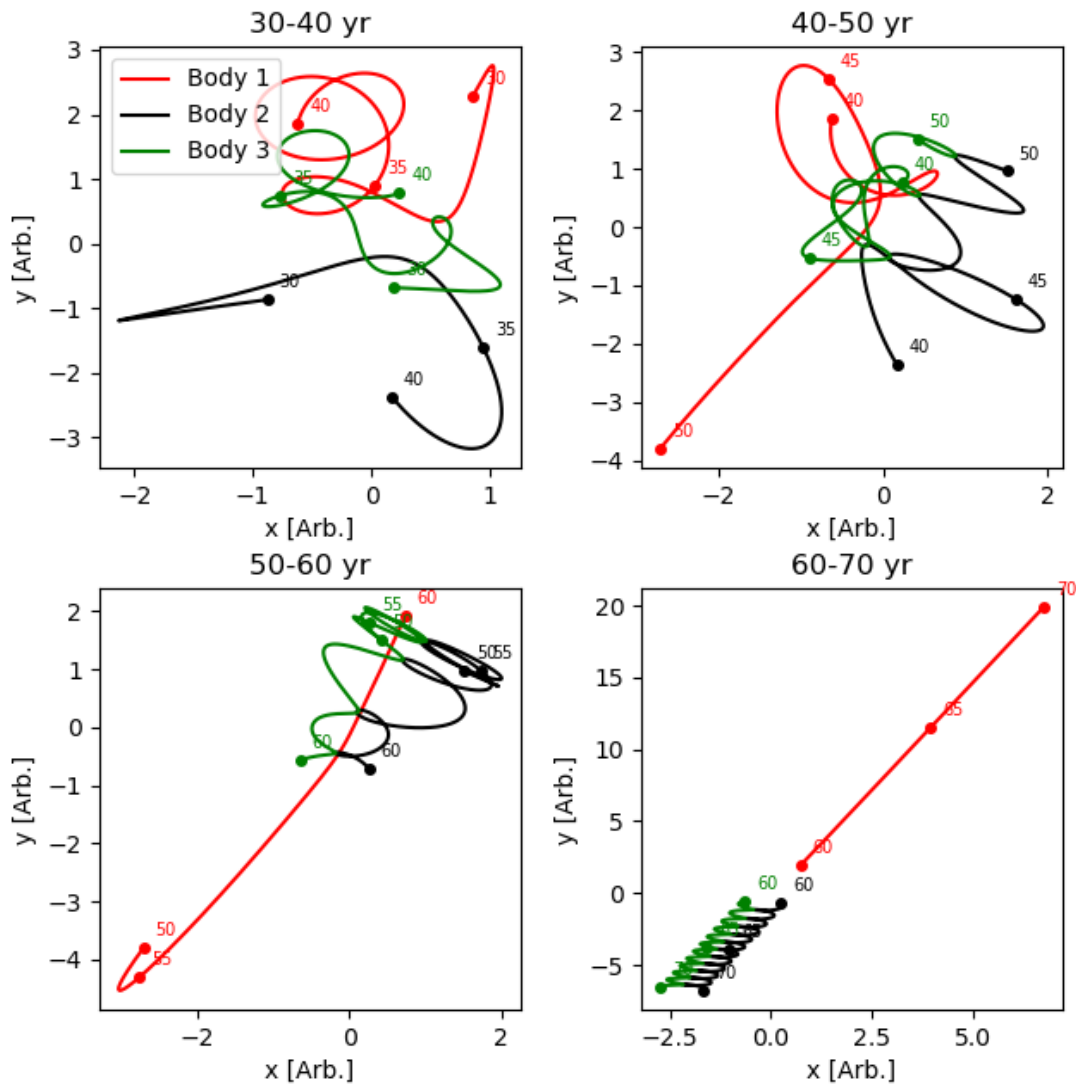


Figure 3: This figure shows the projected trajectories in the x-y plane for Burrau's Pythagorean three body problem, using dimensionless units with $G = 1$, masses $m_1 = 3$ (red), $m_2 = 4$ (black) and $m_3 = 5$ (green). The three bodies start at the vertices of a 3-4-5 right triangle and are initially at rest. The four panels display successive decades of the evolution, from 30–40 yr (top left) to 60–70 yr (bottom right). In each panel the continuous curves give the full orbits in that decade, while dots mark the positions every 5 years. Between year 30 and 50 the motion remains strongly chaotic, with repeated near passages and exchange interactions. By 50–60 yr a stable binary has formed between the m_2 and m_3 bodies traveling in the negative x-y direction. The lighter m_1 body has flung off at escape velocity. These orbits are consistent with the qualitative behaviour reported in the original *Burrau's Problem of Three Bodies* (1967).

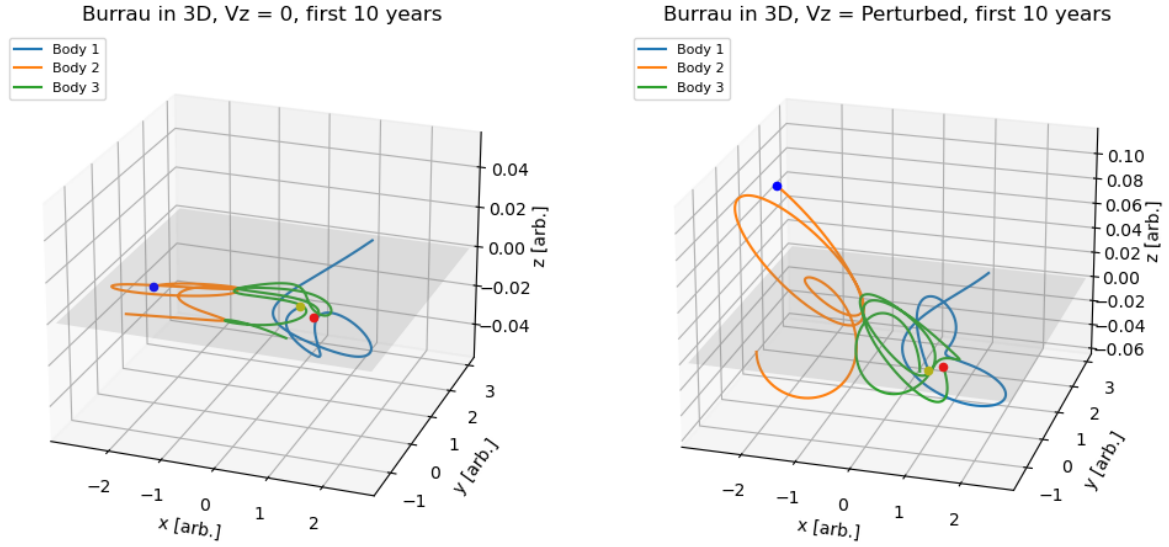


Figure 4: Burrau’s three-body problem integrated in three dimensions to validate the 3D extension of the N-body solver. The initial conditions are the same as before (Pythagorean 3-4-5 triangle), however a perturbation in the new Z direction is applied in the right figure. The trajectories shown are the first 10 years only (to avoid visual clutter during the early chaotic phase). Left: $v_{z1} = v_{z2} = v_{z3} = 0$, the motion remains confined to the $z = 0$ plane (Marked with grey planar sheet), as expected for planar initial conditions. Right: a controlled out-of-plane perturbation $v_{z1} = 0$, $v_{z2} = -0.03$, $v_{z3} = +0.03$ produces a 3D orbit evolution, demonstrating that the code correctly handles fully three-dimensional dynamics. With a velocity perturbation applied to two of the bodies in the z-axis, all three bodies subsequently develop non-zero motion along the z-axis

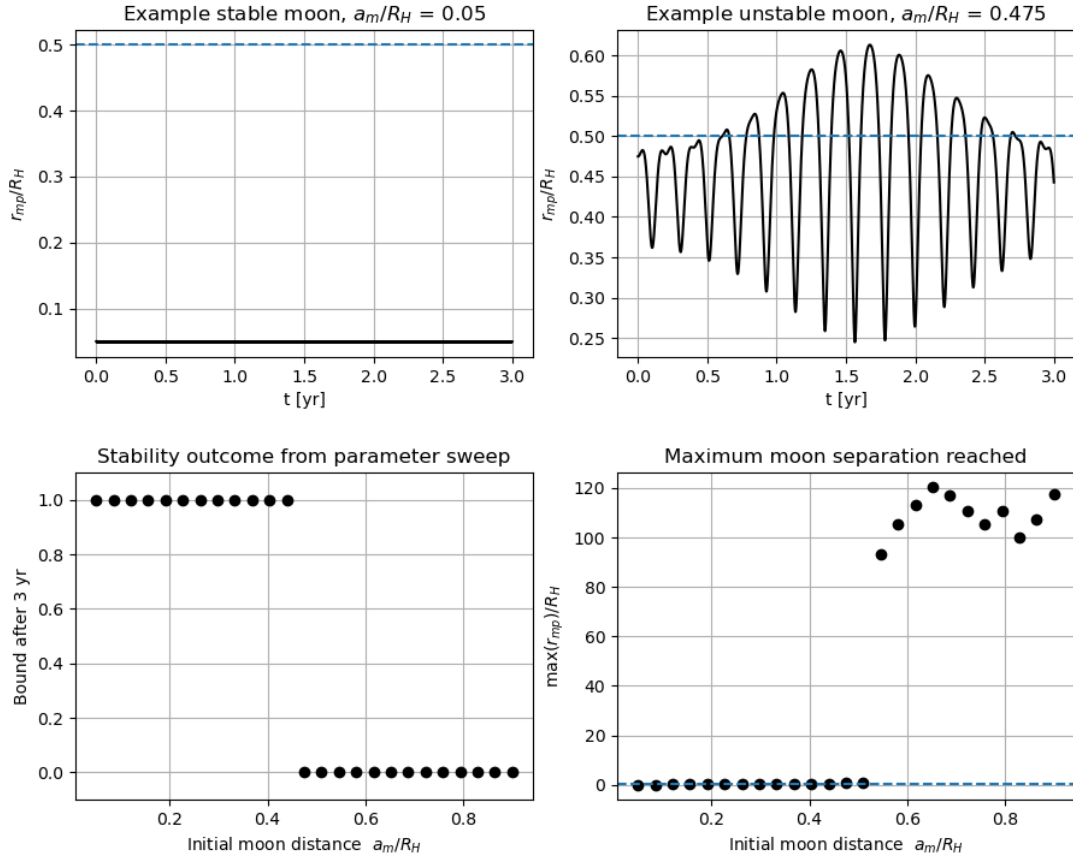


Figure 5: Hill-radius stability test for a moon orbiting an Earth-mass planet around a Solar-mass star at $a_p = 1$ AU (full 3-body gravity solved with `solve_ivp`, planar initial conditions). The moon is launched at a range of initial separations a_m , shown in units of the planet's Hill radius $R_H = a_p(m_p/3M_{star})^{1/3}$. For each run, the moon-planet separation $r_{mp}(t)$ is tracked for 3 years, and the system is classed as bound if $\max(r_{mp}) < 0.5 R_H$ (blue dashed line). Top panels show r_{mp}/R_H for an example stable case ($a_m/R_H = 0.05$) and an example unstable case ($a_m/R_H = 0.475$) which crosses the threshold. Bottom left shows the bound/escape outcome across the parameter sweep, giving a clear stability boundary close to $a_m/R_H \approx 0.5$. Bottom right shows $\max(r_{mp})/R_H$, where unstable cases reach separations $\gg R_H$, consistent with escape from the planet's Hill sphere.

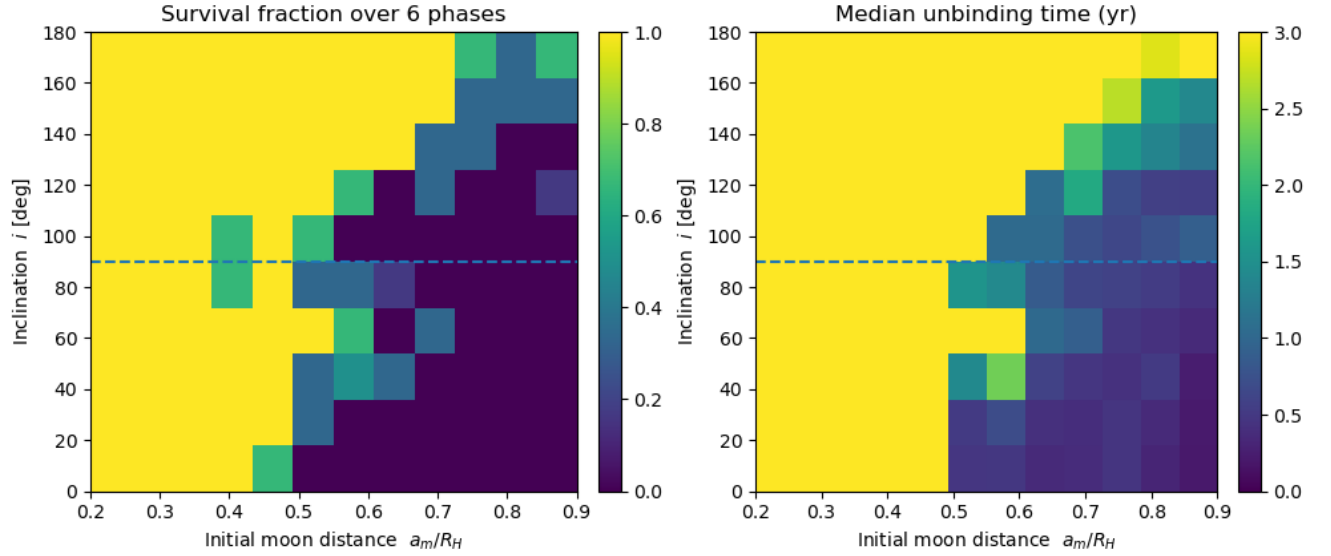


Figure 6: Stability of a moon in a Sun-planet-moon three-body model, mapped as a function of initial moon distance and orbital inclination. The planet is placed on a circular orbit at $a_p = 1$ AU about a solar-mass star, and the moon is launched at $a_m/R_H = [0.2, 0.9]$, where $R_H = a_p (m_p/3m_{star})^{1/3}$ is the planet Hill radius. For each $(a_m/R_H, i)$ point, six initial moon orbital phases are integrated for 3 years using the same `solve_ivp`, and the moon is classed as “unbound” when its planet-centred orbital energy becomes positive. Left: survival fraction (0–1) over the six phases. Right: median unbinding time (years). The dashed line marks $i = 90^\circ$, separating prograde and retrograde cases, with retrograde orbits remaining stable to larger a_m/R_H .

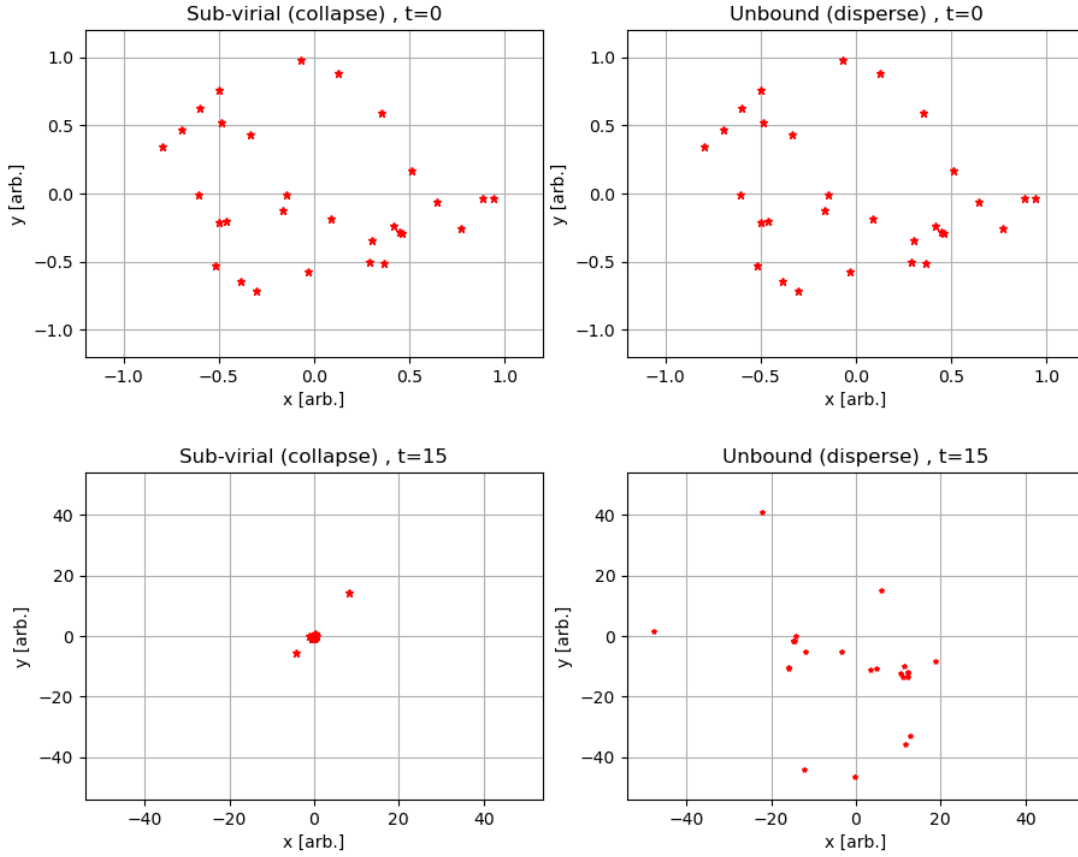


Figure 7: Evolution of an equal mass stellar cluster using the `solve_ivp`. $N = 30$ stars (all $m = 1$) are initialised with random positions inside a sphere of radius $R_0 = 1$ (arb. units), then shifted so the centre of mass position and total momentum are zero. Gravitational forces use a softening length $\epsilon = 0.15$ (arb.) to avoid numerical blow ups during close encounters. The initial velocities are scaled to set the energetic state of the cluster: a bound, low-speed case ($Q = 0.3$, left column) and an unbound, high-speed case ($Q = 1.5$, right column). Each panel shows the x - y projection at $t = 0$ (top) and $t = 15$ (bottom). The bound case collapses into a dense core with a few escapers, while the unbound case disperses with wide spreading and fast outliers.

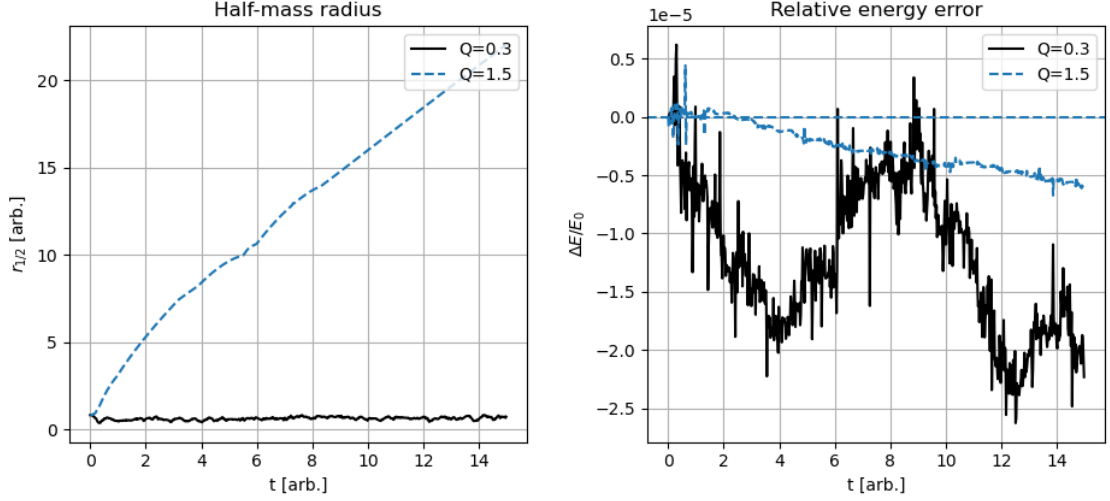


Figure 8: Diagnostics for the equal mass $N = 30$ stellar cluster runs ($G = 1$, $R_0 = 1$, softening length $\epsilon = 0.15$, arbitrary units). Left, half mass radius $r_{1/2}(t)$, defined as the median distance of the stars from the tracked centre of mass (so it tracks the cluster size while ignoring a few extreme escapers). Two initial speed scalings are shown, labelled by $Q = K/|U|$ (Virial ratio) at $t = 0$: the bound case ($Q = 0.3$) remains compact over $t = 15$, while the higher energy unbound case ($Q = 1.5$) expands rapidly, giving a steadily increasing $r_{1/2}$. Right graph is relative total energy error $\Delta E/E_0$ (kinetic plus softened gravitational potential), plotted on a 10^{-5} scale. The energy drift stays small for both runs, indicating the numerical integration is accurate over the time length, Tolerances of $1e^{-6}$ or lower were required in the solve_ivp integral for this accurate energy calculation.